THE TRANSPORT ENERGY OF THE ONE-STEP PRIME-CORRECTED CLOCK

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ABSTRACT. Let τ_k be the midpoint Riemann–von Mangoldt clock, γ_k the ordinates of the zeros of the Riemann zeta function, S_X the phase of the Euler product over the primes $p \le X$, and $\mathfrak{t}_k(X) = S_X(\gamma_k) - S_X(\tau_k)$ the transport of the one-step corrected clock of [5]. We prove that for every fixed $c \in (0, \frac{1}{2}]$ the mean square of $\mathfrak{t}_k(T^c)$ over $T < \gamma_k \le 2T$ equals $\frac{1}{2\pi^2}\log\log\log T + O_c((\log\log\log T)^{5/8})$. Consequently the error of the one-step clock with a fixed cutoff exponent is unbounded in mean square in units of the mean spacing, with a leading constant independent of c; this answers negatively Question 13.2 of [5]. The proof is unconditional. It replaces the zeros by the second prime-corrected clock of [5] and evaluates the remaining correlation at the clock points by an aggregated two-scale Taylor expansion of a Dirichlet polynomial composed with the first corrected clock.

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1. STATEMENT

Notation follows [5]: γ_k are the ordinates of the nontrivial zeros of the Riemann zeta function counted with multiplicity, $N_0(t) = \vartheta(t)/\pi + 1$, τ_k is defined by $N_0(\tau_k) = k - \frac{1}{2}$, $s_k = N_0(\tau_k) - N_0(\gamma_k)$, $\Lambda(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$, $u_k(X) = s_k - S_X(\gamma_k)$, $\mathfrak{t}_k(X) = S_X(\gamma_k) - S_X(\tau_k)$, $\tau_k^X = \tau_k - S_X(\tau_k)/\Lambda(\tau_k)$ and $r_k(X) = \Lambda(\tau_k)(\tau_k^X - \gamma_k)$. Means $\langle\cdot\rangle$ are over $\mathcal{I}_T = \{k : T < \gamma_k \le 2T\}$, $M = \#\mathcal{I}_T$.

Theorem 1.1. *For every fixed $c \in (0, \frac{1}{2}]$, as $T \to \infty$,*

$$B_c(T) := \langle \mathfrak{t}_k(T^c)^2 \rangle = \frac{1}{2\pi^2}\log\log\log T + O_c\big((\log\log\log T)^{5/8}\big), \tag{1.1}$$

and $\langle r_k(T^c)^2\rangle = \frac{1}{2\pi^2}\log\log\log T + O_c\big((\log\log\log T)^{5/8}\big)$.

Date: 23 October 2026.

Corollary 1.2. *For every fixed $c \in (0, \frac{1}{2}]$, $\langle r_k(T^c)^2 \rangle \to \infty$. The one-step clock with a fixed cutoff exponent does not have an error bounded in mean square in units of the mean spacing.*

The upper bound of the same order is Theorem D″ [5, Thm. 7.5]. At $2^{19} < \gamma_k \le 2^{20}$ the defect and the transport compensate to 87–98% [5, Table 7]; Theorem 1.1 shows that this compensation is a finite-height effect. The proof uses a band of primes $T^{\theta_T} < p \le T^c$ with $\theta_T \sqrt{\log\log T} \to \infty$, which is empty unless $\log\log T > 2\pi^2 \omega_T^2 / c^2$ (Section 2).

2. Notation and parameters

Let T be large and $c \in (0, \frac{1}{2}]$ fixed. Put

$$\begin{aligned} &H_0 = \log\log T, \quad H_1 = \log H_0, \quad H_2 = \log H_1, \quad X_1 = T^{H_0^{-1/2}}, \quad X_2 = T^{H_1^{-1/2}}, \\ &V = \frac{H_0}{2\pi^2}, \quad \omega = (\log V)^{1/8}, \quad \theta_T = \frac{\omega}{\sqrt{V}}, \quad Y = T^{\theta_T}, \quad Z = T^{H_0^{-2}}, \quad X = T^c. \end{aligned} \tag{2.1}$$

Since $\theta_T \sqrt{H_0} = \pi\sqrt{2}\,\omega \to \infty$ and $\theta_T \sqrt{H_1} \to 0$, for $T \ge T_0(c)$ we have $Z < X_1 < Y < X_2 < X \le T^{1/2}$. Let

$$L = \frac{1}{2\pi^2} \log \frac{c}{\theta_T} = \frac{H_1}{4\pi^2} + O(\log H_1). \tag{2.2}$$

The phase of the Euler product is [5, eq. (2.10)]

$$S_X(t) = \sum_{p \le X} \sum_{m \ge 1} \beta_{p^m} \sin(mt \log p), \qquad \beta_{p^m} = -\frac{1}{\pi m\, p^{m/2}}, \tag{2.3}$$

and $a_p = \beta_p = -1/(\pi\sqrt{p})$. We write $W = S_X - S_Y$, $Q = S_{X_2}$ and

$$\rho_k = x_k^{(1)} = \tau_k - \frac{S_{X_1}(\tau_k)}{\Lambda(\tau_k)},$$

the first corrected point of [5, eq. (8.1)] with cutoff X_1 (exactly, since $N_0(\tau_k) = k - \frac{1}{2}$). Let $\mathfrak{b}_T = 2\pi/\log(3T)$. For $k \in \mathcal{I}_T$ and $T \ge T_0$, $\tau_k \in [T-1, 2T+1]$, $\tau_{k+1} - \tau_k \ge \mathfrak{b}_T$, and $M \asymp T \log T$ [5].

A finite sum $G(t) = \sum_\lambda g_\lambda e^{i\lambda t}$ over $\lambda \in \{\log(a/b) : a, b \in \mathbb{N}\}$ is called a *rational trigonometric polynomial*; $\|G\|_1 = \sum |g_\lambda|$; its *height* is the largest $\max(a,b)$ with $g_{\log(a/b)} \ne 0$ in reduced form. Its *model mean* is $\mathbb{E}_{\mathrm{mod}} G = g_0$. Equivalently, $\mathbb{E}_{\mathrm{mod}}$ is the expectation when $t \log p$ is replaced by independent uniform phases θ_p for each prime p, and $t \log p^m$ by $m\theta_p$ (unique factorization).

A function Π is *coefficient-dominated by* S_{X_1} if $\Pi = \sum_{p \le X_1} \sum_{m:\, p^m \le X_1^2} \varepsilon_{p,m} \beta_{p^m} \sin(mt \log p) + v E_1(t)$ with $|\varepsilon_{p,m}| \le 1$, $0 \le v \le 1$, and $E_1 = \sum_{p \le X_1} \sum_{m:\, p^m > X_1^2} \beta_{p^m} \sin(mt \log p)$.

3. Results of Paper I used

From [5] we use, for $T \ge T_0$:

- (I1) [5, Thm. 5.1] (Theorem B): for $2 \le X' \le T^{1/2}$ and $c' = \log X'/\log T$, $\langle u_k(X')^2 \rangle = \frac{1}{2\pi^2} \log \frac{1}{c'} + O(1 + \sqrt{\log(1/c')})$.
- (I2) [5, Lemma 7.2] (transport lemma): for $2 \le X' \le T^{1/2}$, $\langle \mathfrak{t}_k(X')^2 \rangle \ll \log\log T\,(1 + \log X')^2/\log^2 T$.
- (I3) [5, Prop. 8.1] (exact recursion) with $j = 2$: $\Lambda(\rho_k) e_{2,k} = u_k(X_2) + S_{X_2}(\gamma_k) - S_{X_2}(\rho_k) + \chi_{2,k}$, $0 \le \chi_{2,k} \le e_{1,k}^2/(4\pi \min(\gamma_k, \rho_k))$.
- (I4) [5, Thm. 8.5] (second Newton step): $\langle \Lambda(\rho_k)^2 e_{2,k}^2 \rangle = \frac{1}{4\pi^2} H_2 + O(1 + \sqrt{H_2})$, and $\langle \chi_{2,k}^2 \rangle \ll T^{-2}$ (in its proof).
- (I5) [5, Prop. 8.17] (almost orthogonality): for $0 < b < 1$ and the vectors $v_p(k) = \sqrt{2/M} \sin(\tau_k \log p)$, $p \le T^b$, the Gram matrix satisfies $\|G - I\|_{\mathrm{op}} \ll_b (\log T)^{-1/4} + T^{b-1} \log T$.

(I6) [5, Lemma 8.3]: the multiplicity function of ρ_k satisfies $\int \tilde{n}^2 \ll \mathfrak{b}_T M$; its proof is reproduced, for general Π, in Lemma 4.5.

External inputs: the Kusmin–Landau inequality [1, Thm. 2.1], the Montgomery–Vaughan mean value theorem [4, Cor. 3], Mertens' theorems [3], and Hoeffding's inequality [2, Thm. 2].

4. Auxiliary lemmas

Lemma 4.1 (Clock sums)**.** *Let $0 < |\omega_0| \le \frac{3}{4}\log T$ and let $(w_k)_{k\in\mathcal{I}_T}$ have $\|w\|_\infty + \operatorname{Var} w \le A$. Then*

$$\Big|\sum_{k\in\mathcal{I}_T} w_k e^{i\omega_0\tau_k}\Big| \ll A\Big(\frac{\log T}{|\omega_0|}+1\Big).$$

In particular, if $\omega_0 = \log(a/b) \neq 0$ with $a, b \le T^\beta$, $\beta \le \frac{3}{4}$, then $\frac{1}{M}|\sum_k w_k e^{i\omega_0\tau_k}| \ll A\,T^{\beta-1}$.

Proof. Take $\omega_0 > 0$. Let $\Omega(x) = \omega_0\tau_x/2\pi$ for real x. By [5, eq. (2.5)], $\Omega'(x) = \omega_0/\log(\tau_x/2\pi)$, continuous and monotone, and on the window $0 < \Omega' \le \frac{3}{4}\log T/\log((T-1)/2\pi) \le 0.8$ for $T \ge T_0$. Hence $\|\Omega'\| \ge \min(\omega_0/\log(3T), \frac{1}{5})$ and by Kusmin–Landau every partial sum of $e^{i\omega_0\tau_k}$ over consecutive $k \in \mathcal{I}_T$ is $\ll \log T/\omega_0 + 1$. Abel summation gives the weighted bound. For the last claim, $|\log(a/b)| \ge |a-b|/\max(a,b) \ge T^{-\beta}$ and $M \asymp T\log T$. □

Lemma 4.2 (Clock means versus the model)**.** *Let G be a rational trigonometric polynomial of height $\le T^\beta$, $\beta \le \frac{3}{4}$, and (w_k) as in Lemma 4.1. Then $|\frac{1}{M}\sum_k w_k G(\tau_k) - \mathbb{E}_{\mathrm{mod}} G \cdot \frac{1}{M}\sum_k w_k| \ll A\,\|G\|_1 T^{\beta-1}$.*

Proof. Apply Lemma 4.1 to each $\lambda \neq 0$ and sum. □

Lemma 4.3 (Sub-Gaussian model)**.** *Let $P = \sum_{p\in\mathcal{P}} f_p(\theta_p)$ with θ_p independent and uniform, $f_p(\theta) = \sum_m b_{p,m}\sin(m\theta)$ finite and $|f_p| \le c_p$. With $\sigma^2 = \sum_p c_p^2$, for every integer $q \ge 1$,*

$$\mathbb{E}_{\mathrm{mod}}|P|^{2q} \le 2\,q!\,(2\sigma^2)^q. \tag{4.1}$$

Proof. The $f_p(\theta_p)$ are independent, of mean zero and with values in $[-c_p, c_p]$. Hoeffding's inequality gives $\mathbb{P}(|P| \ge x) \le 2e^{-x^2/2\sigma^2}$, and $\mathbb{E}|P|^{2q} = \int_0^\infty 2qx^{2q-1}\mathbb{P}(|P| > x)dx \le 4q\cdot\frac{1}{2}(2\sigma^2)^q\Gamma(q)$. □

For Π as in (2.3) restricted to a set of prime powers, we use the proxy $\sigma^2(\Pi) = \sum_p(\sum_m |\mathrm{coeff}_{p^m}|)^2$. For the sums of this paper, by Mertens,

$$\sigma^2(S_R) \le \frac{1}{\pi^2}\big(\log\log R + O(1)\big), \qquad \sigma^2(S_{R_2} - S_{R_1}) \le \frac{1}{\pi^2}\Big(\log\frac{\log R_2}{\log R_1} + O\Big(\frac{1}{\sqrt{R_1}}\Big)\Big), \tag{4.2}$$

since $\sum_m |\beta_{p^m}| \le \frac{1}{\pi}(p^{-1/2} + 2p^{-1})$.

Lemma 4.4 (Time means versus the model)**.** *Let G be a rational trigonometric polynomial of height $\le N_G$ and J an interval. Then $|\frac{1}{|J|}\int_J G - \mathbb{E}_{\mathrm{mod}} G| \le 2\|G\|_1 N_G/|J|$.*

Proof. For $\lambda \neq 0$, $|\int_J e^{i\lambda t}dt| \le 2/|\lambda| \le 2N_G$. □

Lemma 4.5 (Multiplicity of shifted clock points)**.** *Let Π be coefficient-dominated by S_{X_1}, $y_k = \Pi(\tau_k)/\Lambda(\tau_k)$, $\varrho_k = \tau_k - y_k$ and $\tilde{n}(u) = \#\{k \in \mathcal{I}_T : u - \mathfrak{b}_T \le \varrho_k \le u\}$. Then $\varrho_k \in [T - T^{1/2}, 2T + T^{1/2}]$ and $\int_{\mathbb{R}} \tilde{n}^2 \le C\mathfrak{b}_T M$ with C absolute, uniformly in Π.*

Proof. We follow [5, Lemma 8.3]. Three facts are used. (a) $\sup|\Pi| \le \sum_{n\le X_1^2}|\beta_n| + |E_1| \le 2X_1 + 1$, so $|y_k| \le T^{1/2}$. (b) $\Pi' = \sum \varepsilon_{p,m}\beta_{p^m} m\log p\cos(mt\log p) + vE_1'$, with $|E_1'| \le \sum_{p\le X_1}\sum_{p^m > X_1^2}\log p\,p^{-m/2} \le 7$. By Montgomery–Vaughan (length $X_1^2 \le T$) and Mertens, $\int_{T-1}^{2T+1}|\Pi'|^2 \ll T\sum_{p\le X_1}\frac{\log^2 p}{p} + T \ll T(1+\log X_1)^2$. (c) $\langle\Pi(\tau_k)^4\rangle \ll (1+H_0)^2$. The polynomial part has height $\le X_1^8 \le T^{1/8}$ in the fourth power, and $\|\cdot\|_1 \le (2X_1)^4$; Lemmas 4.2 and 4.3 with (4.2) give it, and $|E_1| \le 1$.

Now write $b = \mathfrak{b}_T$. The ordered pairs (j,k) with $|j-k| \le 1$ number $\le 3M$. Let $j = k+m$ with $m \ge 2$ and $|\varrho_{k+m} - \varrho_k| \le b$. Since $\tau_{k+m} - \tau_k \ge mb$, $|y_{k+m} - y_k| \ge (m-1)b$.

Variation. On $J_{k,m}=[\tau_k,\tau_{k+m}]$, of length $\le mb(1+o(1))$, $y=\Pi/\Lambda$ has $y'=\Pi'/\Lambda-\Pi\Lambda'/\Lambda^2$. By (a), the second part contributes $\ll T^{1/2}\cdot mb/(T\log^2 T)=o((m-1)b)$. Hence $|J_{k,m}|^{-1}\int_{J_{k,m}}|\Pi'|\ge \frac{1}{3}\Lambda(T)$. Each t lies in at most m of the $J_{k,m}$, and $|J_{k,m}|\ge mb$. By Chebyshev, Jensen and (b),

$$\#\{k:\text{variation}\}\le \frac{9}{\Lambda(T)^2 b}\int_{T-1}^{2T+1}|\Pi'|^2\ll \frac{T(1+\log X_1)^2}{\log T}\ll M\,H_0^{-1}.$$

Size. $\max(|y_k|,|y_{k+m}|)\ge \frac{m-1}{2}b$, so $\max(|\Pi(\tau_k)|,|\Pi(\tau_{k+m})|)\ge\frac{m-1}{3}$. By Chebyshev of order 4 and (c), $\#\{k:\text{size}\}\ll M(1+H_0)^2/(m-1)^4$.

Sum. Summing $M\min(H_0^{-1},H_0^2(m-1)^{-4})$ over $m\ge 2$ gives $\ll MH_0^{-1/4}$. Each pair contributes at most b to $\int\tilde n^2=\sum_{j,k}(b-|\varrho_j-\varrho_k|)_+$. Hence $\int\tilde n^2\le(3+O(H_0^{-1/4}))bM$. □

Lemma 4.6 (Sampling at points of bounded multiplicity)**.** *Let $f(t)=\sum_{n\le N}(\alpha_n\cos(t\log n)+\alpha'_n\sin(t\log n))$ be real, $q\in\{1,2,3,4\}$, $(2\|f\|_1)^{4q}N^{8q}\le T^{1/2}$, and $\varrho_k\in[T-T^{1/2},2T+T^{1/2}]$ with $\int\tilde n^2\le\kappa\mathfrak{b}_T M$, $\kappa\ge1$. Let $\sigma_f^2=\sum_p(\sum_m(|\alpha_{p^m}|+|\alpha'_{p^m}|))^2$. Then*

$$\langle f(\varrho_k)^{2q}\rangle\ll_q \kappa^{1/2}\sigma_f^{2q}(1+\mathfrak{b}_T\log N).$$

Proof. Let $F=f^{2q}\ge0$ and $b=\mathfrak{b}_T$. For $x\in[u-b,u]$, $F(x)\le\frac1b\int_{u-b}^u F+\int_{u-b}^u|F'|$. Since $\varrho_k\in[u-b,u]$ exactly when $u\in[\varrho_k,\varrho_k+b]$, averaging over u and summing over k gives

$$\sum_k F(\varrho_k)\le\frac1b\int\tilde n(u)\big(AF(u)+bA|F'|(u)\big)du,\qquad Ah(u)=\frac1b\int_{u-b}^u h.$$

A has norm ≤1 on L^2. Let $J=[T-2T^{1/2},2T+2T^{1/2}]$; by Cauchy–Schwarz, $\sum_k F(\varrho_k)\le b^{-1}(\kappa bM)^{1/2}(\|F\|_{L^2(J)}+b\|F'\|_{L^2(J)})$.

Here f^{4q} is a rational trigonometric polynomial of height $\le N^{4q}$ with $\|f^{4q}\|_1\le(2\|f\|_1)^{4q}$, so Lemma 4.4 and Lemma 4.3 give $\frac{1}{|J|}\int_J f^{4q}\le 2(2q)!(2\sigma_f^2)^{2q}+o(1)$. The derivative f' is of the same form with coefficients multiplied by $\log n\le\log N$, so $\frac{1}{|J|}\int_J|f'|^{4q}\ll_q(\log N)^{4q}\sigma_f^{4q}$. By Hölder, $\|F'\|_2=2q\|f^{2q-1}f'\|_2\le 2q(\int_J f^{4q})^{(2q-1)/4q}(\int_J|f'|^{4q})^{1/4q}$. Divide by M and use $|J|/(bM)\asymp1$. □

Lemma 4.7 (Clock mean squares)**.** *Write $W=W_{\rm pr}+W_{\rm pp}$ with $W_{\rm pr}=\sum_{Y<p\le X}a_p\sin(t\log p)$. Then*

$$\langle W_{\rm pr}(\tau)^2\rangle=L+o(1),\quad \langle W_{\rm pp}(\tau)^2\rangle\ll Y^{-1/2},\quad \langle W(\tau)^2\rangle=L+o(1),\quad \langle S_Y(\tau)W(\tau)\rangle=O(\sqrt{L}).$$

Proof. By (I5) with $b=c=\frac12$, $\langle W_{\rm pr}(\tau)^2\rangle=\frac12 a_W^{\top}Ga_W=\frac12|a_W|^2(1+O((\log T)^{-1/4}))$, where $a_W=(a_p)_{Y<p\le X}$ and $\frac12|a_W|^2=\frac{1}{2\pi^2}\sum_{Y<p\le X}\frac1p=L+O(1/\log Y)$.

For a real sum g of the form (2.3) with coefficients g_n, Sobolev on the disjoint intervals $\mathcal{U}_k=[\tau_k,\tau_k+\mathfrak{b}_T]\subset[T-1,2T+2]$ [5, proof of Lemma 5.11] gives

$$\langle g(\tau)^2\rangle\le\frac{2}{\mathfrak{b}_T M}\int_{T-1}^{2T+2}g^2+\frac{2\mathfrak{b}_T}{M}\int_{T-1}^{2T+2}g'^2.$$

For $W_{\rm pp}$ truncate $m\le 4\log T$; the tail is $\le 4X2^{-2\log T}$ uniformly. Montgomery–Vaughan gives $\int W_{\rm pp}^2\ll T\sum_n g_n^2+\sum_n g_n^2 n\ll T/Y+X$, and similarly $\int W'^2_{\rm pp}\ll(T/Y+X)\log^4 T$. With $\mathfrak{b}_T M\asymp T$ and $\mathfrak{b}_T\asymp1/\log T$ this gives $\ll Y^{-1}+X\log^4 T/T\ll Y^{-1/2}$.

Then $\langle W^2\rangle=\langle W_{\rm pr}^2\rangle+O(\sqrt{L/Y})+O(Y^{-1/2})$.

Finally $S_Y=S_{Y,\rm pr}+S_{Y,\rm pp}$. By (I5), $\langle S_{Y,\rm pr}(\tau)W_{\rm pr}(\tau)\rangle=\frac12 a_Y^{\top}(G-I)a_W\ll(\log T)^{-1/4}\sqrt{H_0L}=o(1)$, since the two sets of primes are disjoint. The same Sobolev and Montgomery–Vaughan bound gives $\langle S_{Y,\rm pp}(\tau)^2\rangle=O(1)$ and $\langle S_Y(\tau)^2\rangle\ll H_0$. Hence $\langle S_{Y,\rm pp}W_{\rm pr}\rangle=O(\sqrt{L})$ and $\langle S_Y W_{\rm pp}\rangle\ll\sqrt{H_0}Y^{-1/4}$. □

5. Step A: reduction to a linear response

Proposition 5.1. $B_c(T) = 2L - 2\langle s_k W(\tau_k)\rangle + O(H_1^{5/8})$.

Proof. Write $U_k = W(\gamma_k)$ and $V_k = W(\tau_k)$. Since $S_X = S_Y + W$, $\mathfrak{t}_k(X) = \mathfrak{t}_k(Y) + U_k - V_k$; and $U = S_X(\gamma) - S_Y(\gamma) = (s - u(X)) - (s - u(Y)) = u(Y) - u(X)$.

By (I1), $\langle u(X)^2\rangle = O_c(1)$ and $\langle u(Y)^2\rangle = \frac{1}{2\pi^2}\log\frac{1}{\theta_T} + O(1+\sqrt{H_1}) = L + O_c(\sqrt{L})$. Hence $\langle U^2\rangle = L + O_c(\sqrt{L})$.

By (I2) with $X' = Y$, $\langle \mathfrak{t}(Y)^2\rangle \ll H_0(\theta_T \log T)^2/\log^2 T = 2\pi^2\omega^2$. By Lemma 4.7, $\langle V^2\rangle = L + o(1)$.

Expanding the square and bounding the terms with $\mathfrak{t}(Y)$ by Cauchy–Schwarz,

$$B_c = \langle U^2\rangle + \langle V^2\rangle - 2\langle UV\rangle + O(\omega\sqrt{L} + \omega^2).$$

Now $U = s - S_Y(\gamma) - u(X)$ and $S_Y(\gamma) = S_Y(\tau) + \mathfrak{t}(Y)$. So

$$\langle UV\rangle = \langle sV\rangle - \langle S_Y(\tau)V\rangle - \langle \mathfrak{t}(Y)V\rangle - \langle u(X)V\rangle = \langle sV\rangle + O(\sqrt{L} + \omega\sqrt{L}),$$

by Lemma 4.7 and Cauchy–Schwarz. Since $\omega \asymp H_1^{1/8}$ and $L \asymp H_1$, the errors are $O(H_1^{5/8})$. □

6. Step B: removal of the zeros

Proposition 6.1. $\langle s_k W(\tau_k)\rangle = \langle Q(\rho_k)W(\tau_k)\rangle + O(\sqrt{H_1 H_2} + \sqrt{H_1})$.

Proof. By (I3), since $u_k(X_2) = s_k - S_{X_2}(\gamma_k)$,

$$\Lambda(\rho_k)e_{2,k} = s_k - Q(\rho_k) + \chi_{2,k}.$$

By (I4), $\|s - Q(\rho)\|_2 \le \|\Lambda(\rho)e_2\|_2 + \|\chi_2\|_2 \ll 1 + \sqrt{H_2}$. Cauchy–Schwarz with $\|W(\tau)\|_2 \ll \sqrt{L}$ (Lemma 4.7) gives the claim. □

The right side involves only the clock points τ_k and the primes.

7. Step C: the clock correlation

7.1. **Truncations.** Let

- $Q^\flat = \sum_{p\le X_2}\sum_{m:\,p^m\le X_2^2}\beta_{p^m}\sin(mt\log p)$;
- $P_0 = \sum_{p\le Z}\sum_{m:\,p^m\le Z^2}$, $P_1 = \sum_{Z<p\le X_1}\sum_{m:\,p^m\le X_1^2}$, $P_2 = \sum_{p\le Z}\sum_{m:\,Z^2<p^m\le X_1^2}$ (all with terms $\beta_{p^m}\sin(mt\log p)$);
- E_1 as in Section 2.

Then $S_{X_1} = P_0 + P_1 + P_2 + E_1$. Put $\rho_k^\flat = \tau_k - (P_0+P_1)(\tau_k)/\Lambda(\tau_k)$. For $p \le R$, $\sum_{m:\,p^m>R^2} p^{-m/2} \le 3.5/R$, so

$$|Q - Q^\flat| \le \frac{7}{\log X_2}, \qquad |E_1| \le \frac{7}{\log X_1} \qquad \text{uniformly.} \tag{7.1}$$

Lemma 7.1. $\langle Q(\rho_k)W(\tau_k)\rangle = \langle Q^\flat(\rho_k^\flat)W_{\mathrm{pr}}(\tau_k)\rangle + o(1)$.

Proof. (i) $Q \to Q^\flat$. By (7.1) and $\|W(\tau)\|_1 \le \|W(\tau)\|_2$, the cost is $\ll \sqrt{L}/\log X_2$.

(ii) $W \to W_{\mathrm{pr}}$. $\langle Q^\flat(\rho)W_{\mathrm{pp}}(\tau)\rangle \le \|Q^\flat(\rho)\|_2\|W_{\mathrm{pp}}(\tau)\|_2$. By Lemmas 4.5 and 4.6 ($q=1$, $\Pi = S_{X_1}$, $f = Q^\flat$, $N = X_2^2$, $\sigma_f^2 \ll H_0$), $\|Q^\flat(\rho)\|_2 \ll \sqrt{H_0}$. Lemma 4.7 gives $\|W_{\mathrm{pp}}(\tau)\|_2 \ll Y^{-1/4}$.

(iii) $\rho \to \rho^\flat$. With $\Pi_v = P_0 + P_1 + v(P_2 + E_1)$, coefficient-dominated for $v \in [0,1]$,

$$Q^\flat(\rho) - Q^\flat(\rho^\flat) = -\frac{(P_2+E_1)(\tau)}{\Lambda(\tau)}\int_0^1 Q^{\flat\prime}\Big(\tau - \frac{\Pi_v(\tau)}{\Lambda(\tau)}\Big)dv.$$

By Hölder, the cost is $\le \Lambda_{\min}^{-1}\|P_2 + E_1\|_4 \sup_v \|Q^{\flat\prime}(\tau - \Pi_v/\Lambda)\|_4 \|W_{\mathrm{pr}}(\tau)\|_2$. - P_2 has $\sigma^2(P_2) \le \sum_{p\le Z}\sum_{p^m>Z^2} p^{-m} \le 2/Z$ and height $\le X_1^2$, so by Lemmas 4.2 and 4.3, $\|P_2(\tau)\|_4 \ll Z^{-1/2} + T^{-1/8}$;

and $|E_1| \le 7/\log X_1$. - By Lemmas 4.5 and 4.6 ($f = Q^\flat{}'$, coefficients $\le |\beta_n| \log X_2^2$), $\|Q^\flat{}'(\cdot)\|_4 \ll \sqrt{H_0}\log X_2$.

The cost is $\ll \frac{\log X_2}{\log T}\sqrt{H_0}(Z^{-1/2} + \frac{1}{\log X_1})\sqrt{L} \ll \frac{H_0\sqrt{L}}{\sqrt{H_1}\log T} = o(1)$. The condition of Lemma 4.6, $(2\|f\|_1)^{4q}N^{8q} \le T^{1/2}$, holds: $\|Q^\flat\|_1 \le 2X_2 \log X_2^2$ and $N = X_2^2 = T^{2/\sqrt{H_1}}$. □

7.2. **The aggregated expansion.** Let $e_J(z) = \sum_{j<J} z^j/j!$ and $R_J(z) = e^z - e_J(z) = \frac{z^J}{(J-1)!}\int_0^1 (1-u)^{J-1}e^{uz}du$. For a sum $F = \sum_n \phi_n \sin(t\log n)$ write $F^{(m)}$ for its m-th derivative: $F^{(m)}(t) = \sum_n \phi_n (\log n)^m \Im(i^m n^{it})$.

Lemma 7.2. *Let $J_0, J_1 \ge 1$. Then, with P_0, P_1, Λ evaluated at τ_k,*

$$Q^\flat(\rho_k^\flat) = \mathrm{Main}_k + \mathrm{Rem}_{\mathrm{I},k} + \mathrm{Rem}_{\mathrm{II},k},$$

$$\mathrm{Main} = \sum_{j<J_0}\sum_{l<J_1} \frac{(-P_0)^j(-P_1)^l}{j!\,l!\,\Lambda^{j+l}}\, Q^{\flat(j+l)}(\tau), \tag{7.2}$$

$$\mathrm{Rem}_{\mathrm{I}} = \frac{(-P_1)^{J_1}}{(J_1-1)!\,\Lambda^{J_1}}\int_0^1 (1-u)^{J_1-1} Q^{\flat(J_1)}\Big(\tau - \frac{P_0+uP_1}{\Lambda}\Big)du, \tag{7.3}$$

$$\mathrm{Rem}_{\mathrm{II}} = \sum_{l<J_1} \frac{(-P_1)^l(-P_0)^{J_0}}{l!\,(J_0-1)!\,\Lambda^{l+J_0}}\int_0^1 (1-u)^{J_0-1} Q^{\flat(l+J_0)}\Big(\tau - \frac{uP_0}{\Lambda}\Big)du. \tag{7.4}$$

Proof. For each n, with $\xi = \log n/\Lambda(\tau_k)$, $\sin(\rho^\flat \log n) = \Im(n^{i\tau}e^{a+b})$, where $a = -i\xi P_0$ and $b = -i\xi P_1$. The algebraic identity

$$e^{a+b} = e_{J_0}(a)e_{J_1}(b) + e^a R_{J_1}(b) + R_{J_0}(a)e_{J_1}(b)$$

follows from $e^b = e_{J_1}(b) + R_{J_1}(b)$ and $e^a = e_{J_0}(a) + R_{J_0}(a)$. Insert it and sum $\sum_n \beta_n \Im(n^{i\tau}\,\cdot)$ over the prime powers n of $Q^\flat$.

In the first term, $(-i\xi)^{j+l}n^{i\tau} = \Lambda^{-(j+l)}(\log n)^{j+l}(-1)^{j+l}i^{j+l}n^{i\tau}$ gives (7.2), because P_0, P_1, Λ are real.

In the second, $e^a R_{J_1}(b) = \frac{b^{J_1}}{(J_1-1)!}\int_0^1 (1-u)^{J_1-1}e^{a+ub}du$, and $n^{i\tau}e^{a+ub} = n^{i(\tau-(P_0+uP_1)/\Lambda)}$, which gives (7.3). The third is the same and gives (7.4). □

7.3. **The remainders.** Let $\Lambda_{\min} = \Lambda(T-1)$, $\hat\xi = \log(X_2^2)/\Lambda_{\min} = \frac{4\pi}{\sqrt{H_1}}(1 + O(\frac{1}{\log T}))$ and $\tilde Q^{(m)} = Q^{\flat(m)}/(\log X_2^2)^m$; the coefficients of $\tilde Q^{(m)}$ are bounded by $|\beta_n|$. By (4.2),

$$\sigma_0^2 := \sigma^2(P_0) \le \frac{H_0}{\pi^2}(1+o(1)), \qquad \sigma_1^2 := \sigma^2(P_1) \le \frac{1}{\pi^2}\Big(\frac{3}{2}H_1 + O(1)\Big), \tag{7.5}$$

so that $s := \hat\xi^2\sigma_0^2 \le \frac{16H_0}{H_1}(1+o(1))$ and $\hat\xi^2\sigma_1^2 \le 25$ for $T \ge T_0$. Choose

$$J_1 = \lceil H_1 \rceil, \qquad J_0 = \lceil 24e^2 s\rceil \asymp \frac{H_0}{H_1}. \tag{7.6}$$

Lemma 7.3. $\langle(|\mathrm{Rem}_{\mathrm{I}}| + |\mathrm{Rem}_{\mathrm{II}}|)\,|W_{\mathrm{pr}}(\tau)|\rangle \ll H_0^{1/2-(\log H_1)/2+O(1)}\sqrt{L} + 2^{-J_0}H_0^{1/2}\sqrt{L} = o(1)$.

Proof. Moments of P_0, P_1 at the clock. For integers q with $(2Z)^{2q}Z^{4q} \le T^{1/4}$, which holds for $q \le H_0^2/40$, Lemma 4.2 (height $\le Z^{4q}$, $\|P_0^{2q}\|_1 \le (2Z)^{2q}$) and Lemma 4.3 give $\langle P_0(\tau)^{2q}\rangle \le 2q!(2\sigma_0^2)^q + T^{-1/2}$. The same holds for P_1 with $q \le \sqrt{H_0}/40$.

$\mathrm{Rem}_{\mathrm{I}}$. With the probability measure $\mu(du) = J_1(1-u)^{J_1-1}du$, (7.3) gives

$$|\mathrm{Rem}_{\mathrm{I}}| \le \frac{(\hat\xi|P_1|)^{J_1}}{J_1!}\int_0^1 |\tilde Q^{(J_1)}(x_k(u))|\mu(du), \qquad x_k(u) = \tau_k - \frac{(P_0+uP_1)(\tau_k)}{\Lambda(\tau_k)}.$$

By Hölder ($\frac14+\frac14+\frac12=1$) and uniformly in u: - By the moments above with $q=2J_1$, $\|(\hat\xi P_1)^{J_1}/J_1!\|_4 \le (2(2J_1)!)^{1/4}(50)^{J_1/2}/J_1! \le 2(100e^2/J_1)^{J_1/2}$. - $\Pi = P_0+uP_1$ is coefficient-dominated. Lemmas 4.5 and 4.6 ($q=2$, $N=X_2^2$) give $\|\tilde Q^{(J_1)}(x(u))\|_4 \ll \sqrt{H_0}$. - With $\|W_{\rm pr}\|_2 \ll \sqrt{L}$ and $J_1=\lceil H_1\rceil$, the product is $\ll \sqrt{H_0}(740/H_1)^{H_1/2}\sqrt{L}$, and $(740/H_1)^{H_1/2}=H_0^{-(\log H_1)/2+O(1)}$.

$\mathrm{Rem_{II}}$. Similarly, with $x_k'(u)=\tau_k-uP_0(\tau_k)/\Lambda(\tau_k)$, Hölder with $\frac18+\frac18+\frac14+\frac12=1$ gives

$$\langle|\mathrm{Rem_{II}}||W_{\rm pr}|\rangle \le \sum_{l<J_1}\Big\|\frac{(\hat\xi P_1)^l}{l!}\Big\|_8\Big\|\frac{(\hat\xi P_0)^{J_0}}{J_0!}\Big\|_8 \sup_u\|\tilde Q^{(l+J_0)}(x'(u))\|_4\|W_{\rm pr}\|_2.$$

- $\|(\hat\xi P_1)^l/l!\|_8 \le 2(200e^2/l)^{l/2}$, whose sum over l is $O(1)$. - $\|(\hat\xi P_0)^{J_0}/J_0!\|_8 \le (2(4J_0)!)^{1/8}(2s)^{J_0/2}/J_0! \le 2(8e^2s/J_0)^{J_0/2}\le 2^{1-J_0}$ by (7.6). The moments of P_0 of order $8J_0 \ll H_0/H_1$ are within the range $q\le H_0^2/40$. - The third factor is $\ll\sqrt{H_0}$ as before. □

7.4. **The main term.**

Lemma 7.4 (Non-zero frequencies)**.** *Write* $\mathrm{Main}\cdot W_{\rm pr}=\sum_{j,l}w_k^{(j+l)}G_{j,l}(\tau_k)$ *with* $w_k^{(m)}=\Lambda(\tau_k)^{-m}$ *and* $G_{j,l}=\frac{(-P_0)^j(-P_1)^l}{j!l!}Q^{\flat(j+l)}W_{\rm pr}$. *Then*

$$\sum_{j<J_0,\,l<J_1}\Big|\frac1M\sum_k w_k^{(j+l)}\big(G_{j,l}(\tau_k)-\mathbb{E}_{\rm mod}G_{j,l}\big)\Big| \ll T^{3c/2-1+o(1)}.$$

Proof. $G_{j,l}$ is a rational trigonometric polynomial. Its height is

$$\le Z^{2J_0}X_1^{2J_1}X_2^2X = T^{c+\eta_T},\qquad \eta_T=\frac{2J_0}{H_0^2}+\frac{2J_1}{\sqrt{H_0}}+\frac{2}{\sqrt{H_1}}\to0,$$

and $\|G_{j,l}\|_1 \le \frac{(2Z)^j(2X_1)^l}{j!l!}(\log X_2^2)^{j+l}2X_2\sqrt{X}$. The weight $w^{(m)}$ is monotone with $\|w^{(m)}\|_\infty+\operatorname{Var}w^{(m)}\le 2\Lambda_{\min}^{-m}$. By Lemma 4.2 with $\beta=c+\eta_T\le\frac34$, each term is $\ll\Lambda_{\min}^{-(j+l)}\|G_{j,l}\|_1T^{c+\eta_T-1}$. The sum over $j<J_0$, $l<J_1$ is at most $J_0J_1(2Z\hat\xi)^{J_0}(2X_1\hat\xi)^{J_1}\cdot 2X_2\sqrt{X}\cdot T^{c+\eta_T-1}=T^{3c/2-1+o(1)}$, since $(2Z\hat\xi)^{J_0}=T^{o(1)}$, $(2X_1\hat\xi)^{J_1}=T^{o(1)}$ and $X_2=T^{o(1)}$. □

Lemma 7.5 (Zero frequency)**.** *Let* $\xi_{p,k}=\log p/\Lambda(\tau_k)$. *Then*

$$\mathrm{Diag}:=\sum_{j,l}\frac1M\sum_k w_k^{(j+l)}\mathbb{E}_{\rm mod}G_{j,l}=\sum_{Y<p\le X_2}\frac{a_p^2}{2}\Big\langle\Re\,\mathbb{E}_{\rm mod}\big[e_{J_0}(-i\xi_{p,k}P_0)\big]\,\mathbb{E}_{\rm mod}\big[e_{J_1}(-i\xi_{p,k}P_1)\big]\Big\rangle_k,$$

and $|\mathrm{Diag}|\ll L\,e^{-2\pi^2\omega^2(1+o(1))}+L\,2^{-J_0}$.

Proof. In the model, $W_{\rm pr}=\sum_{Y<p\le X}a_p\sin\theta_p$, the phases of P_0,P_1 involve only primes $\le X_1<Y$, and $Q^{\flat(m)}=\sum_n\beta_n(\log n)^m\Im(i^me^{i\theta_n})$ with $\theta_{q^e}=e\theta_q$. Fix $p\in(Y,X]$. - A term with $n=q^e$, $q\ne p$, is independent of θ_p and has $\mathbb{E}\sin\theta_p=0$. - If $n=p^e$, then $\mathbb{E}[\sin\theta_p\,\Im(i^me^{ie\theta_p})]=\frac12\Re(i^m)\mathbf{1}_{e=1}$.

So only $n=p$ contributes, which requires $p\le X_2$. Its contribution is

$$\frac{a_p\beta_p}{2}(\log p)^m\Re(i^m)\,\mathbb{E}_{\rm mod}\frac{(-P_0)^j(-P_1)^l}{j!\,l!},$$

with $m=j+l$. Multiplied by $w_k^{(m)}=\Lambda(\tau_k)^{-m}$ this is $\frac{a_p^2}{2}\Re\,\mathbb{E}_{\rm mod}\frac{(-i\xi_{p,k}P_0)^j(-i\xi_{p,k}P_1)^l}{j!\,l!}$, because $(-i\xi)^m(-1)^m=i^m\xi^m$ and $a_p=\beta_p$. Summing over j,l gives the formula, using that P_0 and P_1 involve disjoint primes and are independent in the model.

For the bound: for imaginary z, $|R_J(z)|\le|z|^J/J!$. Hence $|\mathbb{E}_{\rm mod}e_{J_1}(-i\xi P_1)|\le 1+\mathbb{E}_{\rm mod}(\hat\xi|P_1|)^{J_1}/J_1!\le 2$ and $|\mathbb{E}_{\rm mod}e_{J_0}(-i\xi P_0)|\le|\mathbb{E}_{\rm mod}e^{-i\xi P_0}|+2^{1-J_0}$, as in Lemma 7.3.

Write $P_0 = \sum_{r\le Z} f_r(\theta_r)$. Then $|f_r| \le c_r \le \frac{1}{\pi}(r^{-1/2} + 2r^{-1}) \le \frac{3}{\pi\sqrt{r}}$, so $|\xi f_r| \le \hat{\xi}\cdot\frac{3}{\pi\sqrt{2}} < 1$, $\mathbb{E}f_r = 0$ and $\mathbb{E}f_r^2 \ge \frac{1}{2}\beta_r^2 = \frac{1}{2\pi^2 r}$. So

$$|\mathbb{E}e^{-i\xi f_r}| \le 1 - \frac{\xi^2}{4\pi^2 r} + \frac{\xi^3}{6}\Big(\frac{3}{\pi\sqrt{r}}\Big)^3 \le \exp\Big(-\frac{\xi^2}{4\pi^2 r} + \frac{\xi^3}{r^{3/2}}\Big),$$

and by independence and Mertens,

$$|\mathbb{E}_{\mathrm{mod}}e^{-i\xi P_0}| \le \exp\Big(-\frac{\xi^2}{4\pi^2}(H_0 - 2H_1 - C) + C\xi^3\Big).$$

For $Y < p$ and $k \in \mathcal{I}_T$, $\xi_{p,k} \ge \log Y/\Lambda(2T+1) = 2\pi\theta_T(1 + O(1/\log T))$. Hence $\frac{\xi^2}{4\pi^2}H_0 \ge \theta_T^2 H_0(1 - o(1)) = 2\pi^2\omega^2(1-o(1))$, while $\theta_T^2 H_1 \to 0$ and $\xi^3 \to 0$. Finally $\sum_{Y<p\le X_2}\frac{a_p^2}{2} \le L$. □

Proposition 7.6. $\langle Q(\rho_k)W(\tau_k)\rangle = O\big(L\,e^{-2\pi^2\omega^2(1+o(1))}\big) + o(1)$.

Proof. By Lemma 7.1, Lemma 7.2, Lemma 7.3, and $\langle \mathrm{Main}\cdot W_{\mathrm{pr}}\rangle = \mathrm{Diag} + O(T^{3c/2-1+o(1)})$ (Lemma 7.4), with Lemma 7.5. □

8. Proof of the theorem

By Propositions 5.1, 6.1 and 7.6, and since $\omega^2 = (\log V)^{1/4} \asymp H_1^{1/4}$ makes $L\,e^{-2\pi^2\omega^2} \to 0$,

$$B_c(T) = 2L + O(H_1^{5/8}) + O(\sqrt{H_1 H_2}) = \frac{H_1}{2\pi^2} + O(H_1^{5/8}),$$

by (2.2); this is (1.1), since $H_1 = \log\log\log T$.

For r_k: by [5, eq. (7.1)], $r_k = u_k(X) + \mathfrak{t}_k(X) + \chi_k$ with $0 \le \chi_k \ll 1/T$, and $\langle u(X)^2\rangle = O_c(1)$ by (I1). So $|\|r\|_2 - \|\mathfrak{t}\|_2| = O_c(1)$, and $\langle r^2\rangle = B_c + O_c(\sqrt{B_c})$. Corollary 1.2 follows. □

9. Scope and dependence on Paper I

The proof is unconditional. It uses the results (I1)–(I6) of [5] as stated there, and classical results (Kusmin–Landau, Montgomery–Vaughan, Mertens, Hoeffding). Every other step is proved above; Lemma 4.5 reproduces the proof of [5, Lemma 8.3] for coefficient-dominated Π. The implied constants are not explicit; they depend on those of [5].

The method: Step B uses [5, Thm. 8.5] to replace the zero ordinates by the second corrected clock, at a cost $O(\sqrt{H_1 H_2})$. Step C then evaluates a correlation of explicit Dirichlet polynomials at the clock points. The two-scale split $P = P_0 + P_1$ and the order of the expansion in Lemma 7.2 keep the factor $e^{-i\xi P_0}$, whose argument has variance $\asymp H_0/H_1$, unimodular in $\mathrm{Rem_I}$; and summation over n before estimation turns each remainder into one derivative of $Q^\flat$ evaluated at shifted points, controlled by multiplicity. The idea of truncating exponentials of prime sums with moment control goes back to the work of Radziwiłł and Soundararajan on Selberg's central limit theorem [6].

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